

Doc. no.: P2000R4

Date: 2022-10-15

Programming Language C++

Audience: All WG21

Reply to: Roger Orr ([rogero@howzatt.co.uk](mailto:rogero@howzatt.co.uk))

# DIRECTION FOR ISO C++

*H. Hinnant, R. Orr, B. Stroustrup, D. Vandevorde, M. Wong*

## Revision History

- R4: this paper
  - Highlight in red new or significantly changed sections since last revision (P2000r3).
  - Added new Section 5.1 Towards a Safer C++
  - Updated Section 7.6 Balance of concerns
- R3: 2021-12-21 published
  - Impact of pandemic on C++23 and C++26
  - Safety and Security
  - C and C++ compatibility
- R2: online:
  - updated with suggestions from Dan Raviv
  - our view on ISO C++ in 2020 on balance of engineering choices
  - our view on ISO C++ online and F2F
- R1: Post-Prague:
  - rephrased and improved Section 6 on C++23
  - updated authors list in mailing to include all authors
  - Highlight in red new or significantly changed sections since last revision (P2000r0)
- R0: Feb 2020 (Pre-Prague): P2000R0 is a continuation of the P0939 series of papers.
  - Updated Section 7.x on Rationale vs Design vs Wording

## Table of Content

|                                        |          |
|----------------------------------------|----------|
| <b>1 Abstract</b>                      | <b>2</b> |
| <b>2 History</b>                       | <b>3</b> |
| <b>3 The Direction Group</b>           | <b>3</b> |
| <b>4 Long-term Aims (decades)</b>      | <b>4</b> |
| <b>5 Medium-term Aims (3-10 years)</b> | <b>8</b> |
| 5.1 Learning and Teaching C++          | 8        |

|                                                            |           |
|------------------------------------------------------------|-----------|
| 5.2 Library and Language                                   | 10        |
| 5.3 Concrete suggestions                                   | 11        |
| 5.4 Studies Proposal to further C++ directions             | 12        |
| <b>6 Priorities for C++23 and C++26</b>                    | <b>13</b> |
| <b>7 Process Issues</b>                                    | <b>15</b> |
| 7.1 Trust                                                  | 19        |
| 7.2 Proposal processing                                    | 21        |
| 7.2.1 Guidance on Converging on unified proposals          | 22        |
| 7.2.2 ABI Stability                                        | 23        |
| 7.2.3 Safety and Security                                  | 23        |
| 7.2.4 C/C++ Liaison                                        | 23        |
| 7.3 The role of TSs                                        | 24        |
| 7.4 “Details” vs. Design                                   | 25        |
| 7.5 Rationale, Design, and Wording                         | 26        |
| 7.6 Balance of concerns and engineering approach tradeoffs | 27        |
| 7.7 C++ future online and F2F                              | 28        |
| <b>8 The C++ Programmers’ Bill of Rights</b>               | <b>29</b> |
| 8.1 “The C++ Programmers’ Bill of Rights.”                 | 29        |
| <b>9 Caveat</b>                                            | <b>30</b> |
| <b>10 Acknowledgements</b>                                 | <b>31</b> |
| <b>11 References</b>                                       | <b>31</b> |

## 1 Abstract

This is intended as a document updated as times go by and new facts and opinions emerge. It tries to articulate our aims for C++, the challenges faced, and some concrete suggestions for priorities.

This document differs from most papers by aiming for a global view and placing individual features in context, trying to avoid delving into technical details of individual proposals except where those “details” could affect the language as a whole or its tool environments.

To prevent this document from becoming toothless “pure philosophy,” we propose giving priority to specific areas of development and to specific proposals in those areas.

Suggested improvements are welcome.

## 2 History

The Direction Group (DG) was created in response to calls for more thought about the direction of C++'s evolution (language and standard library) and comments about our self-established processes in WG21. The specific "call to action" was [Operating principles for C++](#) [Winkel,2017] by several Heads of National Delegations.

We see C++ in danger of losing coherency due to proposals based on differing and sometimes mutually contradictory design philosophies and differing stylistic tastes. For that reason, we recommend that you (re)read [Winkel,2017] before proposing a new feature (language or library).

[Winkel,2017] quoted heavily from D&E. However, during the presentation of [C++ Stability, Velocity, and Deployment Plans](#) [Winters,2017] at the June 2017 Toronto meeting to an almost complete WG21, Titus Winters asked for a show of hands of who had read "The Design and Evolution of C++" [Stroustrup,1994]. Only about a quarter of the hands went up. Assuming that another quarter was shy or distracted, that indicates that half of the committee have never read the articulation of key design principles and decisions for C++. Part of the reason for that is that copies of D&E can be hard to find. Consequently, we offered every member of the committee a free copy and ensured that Addison Wesley made D&E available again (in paper and electronic versions).

We strongly recommend that someone who wants to push or oppose a proposal read [Stroustrup,1994] and its related HOPL papers ([Stroustrup,1993], [Stroustrup,2007], and [Stroustrup2020]). Our impression is that many members of WG21 have an inaccurate view of how and why C++ succeeded and some are pushing popular approaches that in the past have led to failure (e.g., by languages following those approaches and losing out to C++). Obviously aims, techniques, and ideals change over decades, progress happens, but it is dangerous to operate in ignorance of past successes and failures.

## 3 The Direction Group

The direction group's (DG) charter is [Sutter,2018]:

Direction group. The direction group is a small by-invitation group of experienced participants who are asked to recommend priorities for WG21. Currently, that group consists of: Howard Hinnant, Roger Orr, Bjarne Stroustrup, Daveed Vandevoorde, and Michael Wong. Their charter includes setting forth a group opinion on:

- Evolution direction (language and library): This includes both language and library topics, and includes both proposals in hand and proposals we do not have but should solicit. The direction group maintains a list of the proposals it considers the most important for the next version of C++ or to otherwise make progress such as in a TS, and the design group chairs use that list to prioritize work at meetings. Typically, work on other topics will occur after there's nothing further left to do at this meeting to advance the listed items.

- Providing an opinion on any specific proposal: This includes whether the proposal should be pursued or not, and if pursued any changes that should be considered. Design group participants are strongly encouraged to give weight to an opinion that the direction group feels strongly enough about to suggest.

We are concerned with both the long-term and the shorter-term direction and handling of proposals, especially where they may intersect multiple WGs. When looking at direction and at individual proposals, we try to consider the interests of the larger C++ community, rather than just the narrower interests of WG21 members.

The direction group mailing list is [direction@lists.isocpp.org](mailto:direction@lists.isocpp.org). Opinions welcome.

We operate similarly to a Board of Directors, with an annually-rotating chairman. Bjarne lost a random number draw (written by Howard) and was therefore our chair for the first year:

- Michael Wong (2020, 2025)
- Daveed Vandevoorde (2021, 2026)
- Roger Orr (2022, 2027)
- Bjarne Stroustrup (2018, 2023)
- Howard Hinnant (2019, 2024)

Former members:

- Beman Dawes. It is with deep sadness that we learned of Beman's passing in early 2020. We wish to remember him not only for his contribution within this group, but also for his invaluable influence on C++ and its community (e.g., through the founding of Boost, the shepherding of standard C++ filesystem support, and his crucial early advocacy of the STL), but most of all for being an extraordinary friend and human being. Just about everyone using C++ today has reason to be grateful to Beman. He is sorely missed.

We only speak as "The Direction Group" when we are in unanimous agreement on a topic.

## 4 Long-term Aims (decades)

Assuming we succeed, many of us will be writing and maintaining C++ in 10 years' time and 20 years' time. What kind of aims would we think reasonable for such a time scale? Obviously, long-term aims cannot change every year.

As a community, we want lots of things for C++, but

- Some we can't get because they are impossible.
- Some we can't get because they are incompatible with something else we have or want to have.
- Some we don't yet know how to do.
- Some we don't have the resources to do.

- Some we don't agree to be good for C++.
- Some we don't agree to be worthwhile for C++.
- Some are not ready for standardization but may be in the future, or may never be ready.

We understand that we can't have everything we want and that some of the things we want take a lot of time and work. Given that, we need to clarify our current long-term goals.

No language can be everything for everybody. Trying to achieve that has killed many efforts. We have to decide what C++ is supposed to be and to become.

What is C++?

- C++ is a language for defining and using light-weight abstractions
- C++ supports building resource-constrained applications and software infrastructure
- C++ supports large-scale software development

How do we want C++ to develop?

- Improve support for large-scale dependable software
- Improve support for high-level concurrency models
- Simplify language use
- Address major sources of dissatisfaction
- Address major sources of errors

We fundamentally need:

- *Stability*: Useful code “lives” for decades
- *Evolution*: The world changes and C++ must change to face new challenges.

There is an inherent tension here. We should be very reluctant to break compatibility. People always want a simpler language, a language without complicating “legacy features”, and get seriously angry if we break code that they depend on.

C++ is complicated, too complicated, yet we cannot remove significant facilities and changing them is very hard. Changing parts deemed insignificant can be risky (we need better analysis tools [Winter,2016]) and the potential gains would be insignificant. However, we badly need to simplify use of C++ and that leaves us with three alternatives:

- Provide simpler alternatives for simple uses
- Provide simplifying generalizations
- Provide alternatives to complicated and/or error-prone features

Often, a significant improvement involves a combination of those three.

Calling C++ a “multi-paradigm language” is no excuse for adding support for incompatible programming styles. If features don’t smoothly interoperate, we get *de facto* dialects. Today, some of the most powerful design techniques combine aspects of traditional object-oriented programming, aspects of generic programming, aspects of functional programming, and some traditional imperative techniques. Such combinations, rather than theoretical purity, are the ideal.

- Provide features that are coherent in style (syntax and semantics) and style of use

This applies to libraries, to language features, and to combinations of the two.

Technically, C++ rests on two pillars:

- A direct map to hardware (initially from C)
- Zero-overhead abstraction in production code (initially from Simula, where it wasn’t zero-overhead)

Depart from those and the language is no longer C++. We should not depart from these principles and fall into the trap of:

- Abandoning the past which can seriously jeopardize compatibility. C++ is and will continue to be heavily used in long-lasting systems.
- Not addressing new challenges such as higher-level concurrency models. Should we fail, developers will need to switch to some other framework to gain the best performance.

Over the long term, we must strengthen those two pillars:

- Better support for modern hardware (e.g., concurrency, GPUs, FPGAs, NUMA architectures, distributed systems, new memory systems)
- More expressive, simpler, and safer abstraction mechanisms (without added overhead)

Many useful abstractions should find their way into the standard library:

- In principle, the C++ standard library can be implemented in C++ plus a few “intrinsic operations” for accessing low level machine facilities.

This is a variant of the “Don’t leave room for a language below C++ (except assembler)” rule of thumb from D&E. Where we depart from this principle, we get long-term problems because the semantics easily drift apart from the rest of the language.

C++ relies critically on static type safety for expressiveness, performance, and safety. The ideal is

- Complete type-safety and resource-safety (no memory corruption and no leaks)

This is achievable without added overhead, in particular without adding a garbage collector, and without restricting expressibility (see [A brief introduction to C++'s model for type- and resource-safety](#) [Stroustrup,2015b] and [C++ Core Guidelines](#) [CG,2015]). We don’t consider a program littered with casts type-safe and we recognize that the ideal of complete type-safety and complete resource-safety cannot be achieved while accepting every legal C++ program (as we must for compatibility), so external tools will

be needed (e.g., static analysis of lifetimes), but improved language and library support can help a lot by saving programmers from having to use verbose or error-prone features.

C++ is now used in more domains than ever. We do not specifically recommend any specific domains, as every domain is important to somebody in the larger C++ community – even if their domain isn't well-represented in the committee. But we do intend to broaden our support for domains that are well-represented by the C++ community, even non-traditional ones, as a high-level aim. However here are some areas of general concerns – often cutting across application domains – that we should not ignore:

- *Safety and security*: reduce vulnerability to intrusion and the ability to exploit intrusions (e.g., prevent unsafe input and eliminate type violations from misuse of free store). Applications include autonomous vehicles, medical, avionics safety, critical reliability systems, etc.
- *Simplification*: make simple things simple to do (and thereby make C++ easier to learn).
- *Interoperability*: improve interoperability with important languages and systems (e.g., Python bindings). Modern C++ facilitates that with standard-layout types, variadic templates, lambdas, and more.
- *Support for demanding applications*: offer domain-specific support for important application areas, such as medical, finance, automotive, and games (e.g., key libraries, such as JSON, flat containers, and pool and stack allocators).
- *All system types and sizes from the small to the very large such as for example Edge, Microcontroller, Embedded, Desktop, Data Centers and Cloud Computing*:
  - *Embedded systems*: make C++ easier to use and more effective for large and small embedded systems, (that does not mean just simplified C-like C++; e.g., see [Quora1] [Quora2] [Saks,2016] [Stroustrup,2018]). In particular, embedded systems programming could be supported by more specific libraries, emphasizing compactness of data and predictability of operations (e.g., "no free store allocation").
  - *Data Centers and Cloud Service Providers*: makes C++ better for very large scale computing, including HPC by providing distributed, network and data placement facilities for new forms of memory and compute balance, as well as improvement to workloads for Simulation, Big Data, and Machine Learning. It also supports high compute density, virtualization, and smart resource management.
- Alternatives for error-prone and unsafe facilities (like **std::variant** as an alternative to unions or pattern matching as an alternative for **std::variant**).

This is not an exhaustive list, but demonstrates high-level aims that bracket our priorities. The medium-term aims will bracket our priorities further with specific proposals that are currently in flight. Many of these goals cannot be met through changes in the standard alone. Some, such as bindings and avoidance of error-prone facilities, require improvements beyond the standard, but the standard can make such improvements easier.

We would like to see the software development tools used for C++ (such as compilers, static analyzers, refactoring tools) significantly improved. Most of this is beyond the scope of WG21, but we should try hard not to make things more difficult, e.g. by significantly increasing compile times or the cost of tool building by adding significant complexity barriers (e.g., by encouraging coding styles increasing the use of macros or brittle SFINAE).

To maximize the likelihood of delivering significant improvements,

- We discourage isolated “cute” proposals.

There is a limit to what the community can absorb in a given period of time and while a large number of new features can excite C++ enthusiasts, their addition inevitably leads to an impression of instability among many developers and among decision makers observing the evolution of C++ from a distance. Also, work on small isolated features distracts from the work on fundamental improvements (e.g., to the type system or the foundations of concurrency support), diverts resources, may complicate further improvements, and requires additions to references and teaching materials. In the years leading up to the 1998 standard, the committee members often reminded themselves of the story of the Vasa, the beautiful 17th century battleship that sank on its maiden voyage because of (among other things) insufficient work on its foundation and excessive late additions [Stroustrup,2018c].

## 5 Medium-term Aims (3-10 years)

Naturally, whatever we focus on in the medium term should reflect the aims, principles, and problems identified for the long term. What can we do to get closer to those ideals over the next few years? What are promising areas of development? What is urgent?

### 5.1 Towards a Safer C++

Recently, there has been a lot of discussion about safety in the context of C++. Given the increased importance of safety and security in the world, we consider improved safety an important priority for the evolution of C++.

What does “safety” mean? The most basic guarantee is type safety: An object is used only according to the type with which it was defined. This implies that a type (or subtype) is not accessed before its constructor has completed or after its destructor has been invoked. It also implies memory safety because access through a dangling pointer or through an out-of-range access could access an object through the interface of an inappropriate type. Most casts are unacceptable as are misuses of unions. A data race would violate the object’s defined interface.

The aim for C++ was always complete type safety. That ideal came from Simula, but for decades it was only a dream. Requirements for C compatibility and performance were in the way. Today, with C++ software being widely deployed in critical applications and with software technology greatly improved, strong safety guarantees are within reach. In the context of C++, static guarantees are the ideal. A program that contains violations of the chosen guarantees must not run. Runtime checks are unavoidable in some cases (e.g., range checks), but delaying checks to runtime and having to decide how to handle such violations is second best.



However, type safety is not the only type of safety. In fact, there is a broad range of safety concerns. What if a type is poorly defined, e.g., so that invariants are easily broken? Signed/unsigned problems? Narrowing conversions? What about violating real-time constraints? How about code that terminates in a context where termination is unacceptable? How about failing to unlock a lock? Type safety is a foundational requirement for many kinds of safety requirements, but it is not the only kind. Guaranteeing perfect type safety is neither necessary nor sufficient for a variety of guarantees.

Safety is an important concern, but it is not the only concern. We need to balance it with concerns of performance, tooling, applicability in large code bases, portability, maintainability, interoperability with other languages, expressiveness of language. To wit, every language has loopholes – if nothing else a possibility to call C – or explicit “trusted”/unvalidated sections in which the safety checks are not applied.

One particular concern for C++ is how to move from the current state of affairs in billions of lines of code to a world where a code base can be as safe as required. We consider an incremental, gradual approach the only viable approach at scale. This implies the need to allow users to select guarantees to apply from a menu of guarantees (a “safety profile”).

Statically proving an arbitrary C++ program safe is impossible for most definitions of “safe.” On the other hand, when the code follows a set of appropriate rules, such validation is possible through a combination of language rules for the features used, libraries, and static analysis (incl., local flow analysis).

The conclusion of this is that traditional language changes (enhancements and restrictions) will not support a sufficiently wide range of needs. Fundamentally, any single definition of safety will be wrong for a significant user population. Instead, we need support from a framework of safety profiles supported by static analysis. The C++ Core Guidelines [CG,2015] and their static analysis is an existence proof of the technical viability of this approach for type safety and other needs.

What can the committee do to support such an approach? Just as we cannot require a specific compiler, we cannot require a specific static analyzer. What seems feasible is to enable the specification of general requirements in the code, leaving the implementation to a variety of analyzers. For example, `[[type_safety]]` at the start of a TU or module would indicate that type safety is required for this code and all code invoked by it. A set of such “profiles” (e.g., for type safety and safe arithmetic) would be required by the standard, but the set of profiles must be open to accommodate needs that are not universal or cannot yet be enforced for a particular code base. The issue of how to combine different kinds and degrees of safety must be addressed. For example, see [Stroustrup,2022].

In addition, we encourage continuing work on language and library facilities that increase various forms of safety. Examples of such from the past include range-for and `std::array`.

## 5.2 Learning and Teaching C++

C++ has a major problem with learning and teaching. This could send the C++ community into decline over the next few years. C++ is expert-friendly, but it cannot be just expert-friendly without losing important streams of new talent. This was the case before C++11 and C++11 reversed that negative trend. Currently, the C++ community is more active than it ever was, but this improvement must be sustained with solid improvements in the language and the standard library. It takes more still to sustain

the enthusiasm, but here we are concerned with what WG21 can do. As individuals, we can encourage better tools and libraries. We suggest

- Establish a “Learning and Teaching” SG [now SG-20]

We say “Learning and Teaching” because many (most?) new C++ programmers learn on their own. We must help them. Teachers (incl. professors) are often short of C++ experience, software development experience, and preparation time. We must help them. To paraphrase D&E: C++ needs to be like a weed, not like a garden rose, in order to flourish. We can’t assume a reliable support system.

C++ is seriously underrepresented in academia and often very poorly taught. It has been conventional to start teaching C++ by first introducing the lowest level and most error-prone facilities. Naturally, that discourages students and increases the time needed to get to what students consider meaningful computing (graphics, networking, mathematics, data analysis, etc.). Often, teachers even go to the extreme of insisting on using a C compiler. If the ultimate aim is to teach C++, that’s like insisting people start learning English by reading Beowulf or the Canterbury Tales in their original early-English language versions. Those are great books, but Early English is incomprehensible to most native Modern-English speakers. In addition to the linguistic difficulties, such ancient sources present cultural conventions and idioms that seem very peculiar today. Instead of C, someone could teach Simula to prepare for learning C++. Why don’t people do that? Because the historical approach to teaching language (natural or programming language) complicates and detracts from the end goal: good code.

Why then do teachers use the C-first approach to teach C++? Part is tradition, curriculum inertia, and ignorance, but part of the reason is that C++ doesn’t offer a smooth path to idiomatic, proper, modern use of C++. It is hard to bypass both the traps of low-level constructs and the complexities of advanced features and teach programming and proper C++ usage from the start.

C++ teaching is mostly stuck in a pre-graphics, pre-web world. This is a serious weakness, potentially fatal over the longer term. There are beginners’ books trying to alleviate that (e.g., [Programming -- Principles and Practice Using C++ \(Second Edition\)](#) [Stroustrup,2014]) but they are a small minority and only partially successful. Teaching C++ to modern students will not be really effective until we can offer

- Simple standard graphics, simple interaction, and simple use of browsers (or GUI).
- Simple mechanisms for packaging, distribution, and installation for libraries and programs

The latter is beyond the current scope of WG21, but it is essential for the continuing success of C++; maybe the new Tools SG (SG-15) can help. If WG21 cannot do something official, maybe its members can help establish widely accepted de facto standards (note that languages and systems competing with C++ tend not to have formal standards).

What in the language and libraries helps learning and teaching?

- The ability to start teaching using only what is supplied by every implementation.

Students cannot be expected to download and install libraries on day #1. Even a requirement to **#include** an appropriate set of standard headers can be a burden. Thus, the technique of providing a training-wheels library in the form of a header or precompiled libraries is at most second best. This is a

major technical reason that courses start with pointers, arrays, C-style strings, unions, etc. rather than **std::string**, **std::vector**, **std::variant**, and algorithms.

- The ability to sequence teaching from the simple and fundamental to the advanced.

The snag is that the simple and fundamental typically involves abstractions. This implies that

- It should be trivial to include what is needed for teaching; modules should help here and we need an “all” module and/or a “foundation” module (see also [Clow,2018]) so that novices don’t have to search a textbook or the web to find out which header to **#include** to use a foundational feature. Modules should help here.
- Errors coming from misuse of standard library facilities should be comprehensible; concepts should help here. All templates in the standard library intended for non-expert use should be “conceptualized.”
- There should be support for preventing use of non-approved and or yet-to-be-taught features; this cannot be standardized, but tools for catching counterproductive styles should be encouraged; the Core Guidelines [CG,2015] is a start.

To avoid such start-up problems, teachers sometimes restrict students to C as the simpler-to-start-using subset of C++. Unfortunately, this typically leads to unnecessary learning problems and delays the introduction of good C++ programming styles.

Students just starting to learn programming are not the only ones who need help with learning and initial use. They are most likely not even the majority of new C++ programmers. Other significant groups needing support are programmers coming to C++ from other languages and people who are professionals in some field other than software development (e.g., physicists or biologists), and just need to use C++ as a tool in their work. We recommend that the Teaching and Learning SG (SG-20) consider:

- Help/support for programmers from non-C++ backgrounds, e.g., “C++ for X programmers” guides and specialized curricula.
- Help/support C++ programmers keeping up-to-date with the improving language and libraries (“C++ for C++ programmers”).
- In the context of graphics support, consider what can be done to allow results of scientific computations, simulations, performance measures, and the like to be simply displayed.

“Simply displayed” could be direct support for histograms (as in the very first 1980 C++ library), tables, and graphs in a standard-library and/or support for writing such data out in standard formats for display using other tools. The intent is to support incidental graphics use, rather than applications where graphics is a primary focus.

## 5.3 Library and Language

During the early stages of the development of the first standard, we articulated some principles for what to include in the standard library, including

- Language support

- Facilities that everybody needs
- Facilities needed for communicating among separately developed libraries

We think these are reasonable criteria, but in practice they didn't have much effect. People worked on and over-elaborated individual components (e.g., `std::string`) that didn't interoperate effectively, showed major stylistic differences (reflecting differing design conventions), and often didn't completely cover what programmers considered a topic. The results were complaints, incomplete adoption, and wide use of commercial alternatives. We were rescued from disaster by Alex Stepanov. The STL was a coherent design that was sufficiently complete for its domain to become widely adopted.

If we can supply a feature as a library, we should do so because it is easier to validate a design through experimentation, the facility will be available to users earlier, and a library is usually easier to specify in isolation. However, a library design should not be an excuse for inelegant interfaces, for irregular interfaces, for stylistic differences from built-in language features, or overelaboration. It is always easy to add another function to a class, so library components have a tendency to bloat. Note

- the trouble we have had from `std::vector` using unsigned subscripts while built-in arrays use signed indices (unsigned arithmetic is modular, leading to correctness, performance, and error-detection problems)
- the dramatic differences in the interfaces to `std::any`, `std::optional`, and `std::variant`

At the time, there were of course reasons for those design choices that looked good to many, but we must aim for coherence:

- among standard-library features
- between the standard-library features and the built-in language features

Or, stated differently:

- No feature should be added without someone explicitly considering its interaction with other features (language and library).

We suggest that WGs (in particular, EWG and LEWG) delegate a “consistency review” to a small ad hoc subgroup (it can be a different group every time) for each feature or set of related features to be done concurrently with advanced work on technical details. In most cases, the group should include relative “novices” to ensure that the feature is not limited to expert use, and the process should include attempts at developing *realistic* use cases to confirm that the feature is usable at all.

- We encourage a discussion of criteria of what should be in the standard library and what should not.

The aim is to be able to discuss proposed new standard-library components in the context of articulated criteria. ([SD-8] and [StdLibGuidelines] are good starting points.)

## 5.4 Concrete suggestions

Here are some more concrete suggestions for what can be done over the next few years to bring C++ closer to its long-term ideals:

- **Pattern matching:** List comprehension, and not just for algebraic types. This could simplify code and strengthen type safety. It would eliminate a lot of ad-hoc selection, eliminate unsafe use of unions, low-level use of variants, and eliminate the visitor pattern (which is an expensive workaround that confuses many).
- **Exception and error returns:** There are still people who argue against all use of exceptions and people who claim that exceptions should be used consistently instead of error codes. We'd like to see a clear articulation of where each alternative is ideal, but until we do that we should be careful about what error-handling strategies we support. To make progress, we need better data on costs and performance to evaluate the — often simplistic and narrowly focused — solutions suggested. Without better data and analysis, we risk hitting the N+1 problem ("the problem is that we can't manage N alternatives, so we add a new and better solution, thus having to deal with N+1 alternatives") [Stroustrup,2018c] [Stroustrup2019a]. Contracts ([GDR,2016], [Berne,2019]) could simplify this by taking care of many nasty cases.
- **Static reflection:** C++'s lack of run-time reflection is an essential strength in some areas (such as small embedded systems), but lack of simple reflection in specific areas have let many to switch to alternative languages. In particular, the lack of ability to generate serializer/deserializer pairs and object layout maps has been painful in many areas.
- **Modern networking:** We need networking urgently, but beyond that we need good simple support for interaction with modern browsers and networking (<http>, <https>).
- **Modern hardware:** We need better support for modern hardware, such as executors/execution context, affinity support in C++ leading to heterogeneous/distributed computing support, SIMD/task blocks, more concurrency data structures, improved atomics/memory model/lock-free data structures support. The challenge is to turn this (incomplete) laundry list into a coherent set of facilities and to introduce them in a manner that leaves each new standard with a coherent subset of our ideal.
- **Simple graphics and interaction:** Command-line options and console I/O, while still useful in various contexts, haven't been the primary way of interacting with software for decades. Although modern interactive applications deal with broadly varying devices, almost all are controlled using an "event-driven" software model and most involve a graphical screen. We think the C++ standard library needs a component providing
  - a graphical canvas with basic geometric drawing, text rendering, and bitmap or vector output primitives,
  - a simple input API for that graphical canvas abstract enough to deal with both "click" and "touch" events, and
  - a simple, highly-abstract "menus and dialogs" system (again, abstract enough to not force surfacing distinctions between devices as disparate as "phones" and "desktop computers").

Such a component will, of course, be "optional" to allow for platforms that don't have the needed I/O facilities. Practical uses of this component should be simple enough to be easily teachable to first-year students early in a course. Performance wouldn't be a prime concern for

such a component as it would be competing with the likes of Python and Excel, rather than the excellent professional C++ graphics tools.

- **Improved Unicode support:** for which SG16 was set up, and which has made significant progress since the original version of this paper was published.
- **Anything from the Priorities for C++20 that didn't make C++20:** see [Priorities for C++20](#).

We feel that work on a database interface is badly needed, but since there seems to be no current work on that in the committee and no critical mass of expertise and interest, we can't put it on this list.

## 5.5 Studies Proposal to further C++ directions

In view of several questions that have emerged from various SGs and WG21, there is a need to support research and studies to answer specific questions about C++. In the past, there used to be a Performance TR which answered questions about C++ performance. This is similar but we propose to have individuals volunteer and supply the research. The Directions Group would like to urge volunteers to come forward for studies in the following area. If interested, please email the DG for exact requirements and issues of concern before proceeding.

- What is the impact of optimization in relation to predictability and security in the hands of most programmers? E.g. memcpy can now be entirely optimized even if naively written, while avoiding the undefined behavior specified by the C and C++ standards.
- What are the differences in various OS platforms relative to what the C++ Standard might recommend, e.g. UTF8 is very popular, but not that common on windows
- What is the actual data on exception handling vs. error code (this is an urgent question of SG14)? E.g., What is the difference in size and/or run-time cost if exception handling is enabled but exceptions are not actually thrown?
- What is the actual compile-time and run-time change (in size and time) with Modules? E.g., this is a prime example where a feature is designed to improve compile time — how successful is it?
- What is the actual compile-time and run-time impact (in size and time) of Contracts? E.g., Do contracts add overhead?

## 6 Priorities for C++23 and C++26

C++20 is out of the door and shapes up to be a major step forward for C++. Consequently, we now concentrate on C++23 and beyond.

Unfortunately, almost as soon as we shipped C++20 in the 2020 Prague meeting, the global CoViD-19 pandemic hit and severely affected our process through the C++23 development cycle. On the plus side, the committee as a whole has made significant progress in adapting our processes for a distributed “remote-meeting” environment, and that has allowed us to advance a number of proposals for C++23. On the minus side, we find that the attendance of virtual meetings is less consistent (several key voices are often missing) and we have heard many mentions of “meeting fatigue” (WG21 meetings often compete with other professional and personal obligations). Some of that presumably explains why the proposals that do make it through the new processes are mostly modest. We therefore suggest we should aim at returning to in-person meetings as soon as practical.

We expect the committee to maintain the 3-year cycle.

Ville Voutilainen's "Bold overall plan for C++23" (P0592R4) remains a good target, but must now be adjusted to move some of the goals to C++26. Note the mention of static reflection and functional-programming-style pattern-matching. We encourage the committee to show restraint in adding features and recognizing that after a major release like C++20, we need time to "polish" the feature set based on experience from massive use, especially as relates to feature interactions (including interactions among language features and standard library components), and that the community needs time to develop effective uses of the total feature set. Thus, we should aim at completing the following for C++23:

- A Modular standard library
- Library support for coroutines
- Executors
- Networking

Also of high priority, are the following directions (realistically, now for C++26):

- Pattern Matching
- Reflection

as well as (not mentioned in P0592R4)

- Further Conceptifying the Standard Library
- Further Range improvements (e.g., application of ranges to parallel algorithms and operations on containers and integration with coroutines)

We are in favor of Contracts, but consider it unclear if a consensus can be established in the time available. Given the heated debates leading up to the withdrawal of contracts from C++20 and the obvious disagreements about the scope and potential uses of contracts, we encourage a cautious approach. The wording withdrawn from a pre-C++20 draft was at least the third attempt to add some form of contracts to C++ (and, depending on how you count, the 5th design). If the choice becomes between a delay until C++26 and another heated debate (implying potential failure for C++23), we would recommend a delay. Of course, if a felicitously uncontroversial consensus were to arise early enough, Contracts in C++23 would still be welcome news. However, it appears that Contracts — a largish feature — has also been impacted by the operational effects of the pandemic, and C++26 now looks like a relatively optimistic aim.

### **Other potential candidates for exploration**

After C++20, we encourage focusing on adding library components in preference to language features. Some candidates are already in SGs. We list, in no particular order, the following as potential candidates worth exploring while we work on C++23 and C++26:

- Audio
- Linear algebra
- Graph data structures
- Tree data structures

- Task graphs
- Automatic differentiation
- Light-weight transactional locks
- A new future and/or a new async
- Statistics
- Array style programming through `mdspan`
- Machine learning support (SG19)
- Better support for C++ ecosystem (SG15)
- Further support for heterogeneous programming
- Graphics (SG13)
- An improved definition of “freestanding” implementations
- Education curriculum (SG20)

This is already an ambitious list, and we may find that it is incomplete. Tackling all of those items potentially means that we might overwhelm the capacity of LWG, LEWG, and various subgroups to process ensuing designs. We may therefore need to start additional subgroups or re-activate existing subgroups (like SG9/Ranges) to improve efficiency, as well as consider a separate pipeline for, e.g., projects that would benefit from being first published through a TS.

Note that we chose these topics not because they are easy, but because we estimate they are good for C++’s future directions.

## 7 Process Issues

We are “a bunch of volunteers.”

- Most are enthusiasts for some aspect of the language or other.
- Few have a global view (geographically, C++ community, C++ usage, C++ Language and standard library).
- Most have a strong interest in only some subset of use, language, or library.
- Most are deeply engaged with a specific form of C++ as part of their daily work.
- Our levels and kinds of relevant computer-science, design, and programming language education vary dramatically.
- Many are clever people attracted to clever solutions.
- Some are devoted to ideas of perfection.
- Many are tool builders.
- Some have full control over their source code whereas others critically rely on code outside their direct control (open-source and/or commercial).
- Some operate in an environment with strong management control, others in environments without a formal management structure (and everything in-between).



This implies that we can't rely on a common vocabulary, a core set of shared values, a common set of basic ideals, or a common understanding of what's a problem. Consequently, we must spend more effort on

- articulating rationales for proposals.
- facilities for the "average programmer," who is seriously underrepresented on the committee.
- facilities aimed more at application builders than at builders of foundational libraries.

Please pay special attention to that last point. We feel that C++'s utility and reputation suffer badly from the committee lacking attention to improvements for relative novices and developers with relatively mundane requirements. Remember:

- Most C++ programmers are not like the members of the committee

We, as a committee, have no mechanism of reward (except accepting someone's proposal) or punishment (except delaying or rejecting someone's proposal). To get something accepted, we need consensus (defined as large majority, but not necessarily unanimity). This has implications on what we can do and how we do it.

- Nothing gets done unless someone cares enough to do it
- A small vocal minority can stop any proposal at any stage of the process.

Currently, C++ is probably as popular as it has ever been and there are definitely more active members of the standards committees than ever before. One effect of this enthusiasm for C++ is to inundate us with a flood of proposals. The sheer volume of proposals leads to fewer proposals getting through the processes to get accepted. Some proposals become warped from the need to gain support in an environment where time to think and present ideas is limited. Many of these proposals come from people who are unacquainted with standardization. We understand that some new members find it hard to accept that

- Progress is less rapid than for their corporate and open-source projects.
- Concerns that are essential to them are not given priority by the committee.
- Committee members don't all understand or accept other members' experience and design principles.
- Standardizing for decades differs dramatically from shipping the next release with the fewest number of bugs.
- There are millions of programmers who use C++ in ways that differ from what they consider normal and reasonable.
- "No bugs" does not imply that a proposal is good.
- "There are bugs" does not imply that a proposal is bad (there is a saying among mathematicians that the way to recognize an important result is by the number of errors in its initial proof – it is relatively simple to provide a flawless proof for a trivial result).

- “Good enough” isn’t always good enough because a second look at the problem might come up with a better, more general, or better integrated solution. We have to consider the long term (decades).
- 90% or more of the work of getting a proposal into the standard is ensuring that it fits smoothly with other facilities (language and standard library).
- Stability/compatibility is an essential feature. **For example, any proposal introducing a silent semantic change is likely to lead to significant objection, so providing good technical justification would be recommended for any such proposal.**

And still some improvements are urgent. However, in the context of the committee “urgent” still implies years of work/delay. Novices are not the only ones who are impatient, but we must try to channel our energies into constructive activities. We encourage members (old and new)

- to get acquainted with [C++’s history and design rules](#)
- to accept or contribute to [our long-term aims for C++](#)

The aim of most members, new or “vintage” is to improve C++ by having their favorite proposal accepted. However, the sheer number of proposals (constantly on the order of a hundred) and the number of papers (running at more than one hundred for each meeting), implies that not even every **good** proposal can be accepted. We encourage everyone to ask themselves

- Is my proposal among the top-20 in terms of long-term benefit to the C++ community as a whole?
- Is my proposal essential for some important application domain?

Push only very gently if the answers are “no,” “sort of,” or “I don’t know” and redirect effort to proposals for which the answer is “Yes!” We encourage the WG chairs to

- Focus on the major high-level and intermediate-level goals
- Articulate the goals as they relate to their WG
- Prioritize proposals based on those goals
- Discourage proposals not in that scope
- Discourage proposals from being re-submitted with only minor changes after rejection (especially if the revised proposal does not include new insights into the problem to be solved).
- Encourage proposers to discuss potential disadvantages and costs of their proposals as well as their benefits. Every new feature comes with a significant cost.

All proposals consume the (limited) committee time and WG21 members should be considering the best *overall* outcome for the future of the language. Hence while small proposals to clean up non-trivial defects are welcome, discussion about these may have lower priority. If such a small proposal proves to be controversial it is probably better to withdraw, or defer, it to avoid preventing progress on more substantive items.

We are a set of interrelated committees currently with about 200 members present at a meeting and more active via the Web. Thus some “design by committee”, or rather “design by committees,” is unavoidable. We need to consciously and systematically try to minimize those effects by building a shared sense of direction.

- We have no shared aims, no shared taste.

This is a major problem, possibly the most dangerous problem we face as a committee. For C++ to succeed, we must overcome that. For starters, we – as individuals, as SGs, as WGs, and as the committee as a whole – must devote more time to develop and articulate common understanding. In particular for every proposal being considered we should do more to emphasize the motivation and explain how it fits in the language, standard library, and common uses that we anticipate.

- The alternative is a dysfunctional committee producing an incoherent language.

We need to be more explicit about

- What general problems we are trying to address
- How a particular proposal serves those articulated aims

It would also be a great improvement if members didn’t spend all of their time in a single WG. Doing that leads to lack of insight and of trust.

- Try to spend at least one day each meeting in a WG that isn’t “your own”
- Try to read several papers from each mailing that is not aimed at “your WG”

Note that if “your” proposal progresses, you’ll have to shepherd it through different WGs, so it is good to have some understanding of how they operate.

In addition, the committee as a whole should try to avoid that

- a single organization or cohesive group gains control of a WG.

WGs must be open to inputs from many sources.

As specified in the ISO and NB charters

- the operations of the committee must be transparent.

It is only natural that proposals are first developed and discussed among a small group of friends or colleagues. However, once a proposal reaches a larger group, discussions should be made public to all members. Furthermore, proposals should be made available as complete papers in plenty of time for the members to consider them before meetings. It is not acceptable to present only an incomplete summary a week or two before a meeting and expect a serious discussion, let alone a binding vote. Similarly, it is unreasonable to expect members to decide on a large detailed proposal with only a few days’ warning; remember that a mailing before a meeting consists of about a hundred papers, including many long ones.

To maximize the likelihood of delivering significant improvements and to increase the predictability of our processes, we discourage

- “Change the World” papers for proposals already in flight

We change the language and standard library by gradually building on previous work or by providing a better alternative to an existing feature. Proposals to completely change direction for a proposal already in process should be treated with suspicion and subjected to at least as much scrutiny as the proposal already in flight has been. Looking at new proposals is more exciting than working out the obscure details of an old proposal, but eventually all successful proposals go through the “polish the last details” stage, so we can’t escape that through new and shiny proposals, just postpone the pain. A proposal for a radical change to a proposal already “in flight” (i.e., in its second or later discussion) should not be allowed to delay the latter unless it comes with a paper with a detailed discussion of design, use, and implementation.

## 7.1 Trust

At the start of the formal standardization efforts (ANSI and ISO), P.J. Plauger and Tom Plum emphasized the need for trust in the process. The desired degree of trust is currently missing, and the consequences are dire.

A WG can spend meetings on a proposal, carefully crafting a compromise, balancing concerns, just to have the next WG spend several meetings reviewing the proposal, demanding changes, and rephrasing the wording with patchy understanding of the design rationale as it evolved over time. In particular

- large feature developments should be discussed in joint EWG+CWG, LEWG+LWG, and/or EWG+LEWG sessions as early as the overall design and the understanding of the underlying implementation alternatives allows.

Finally, after years of process, someone then stands up in full committee and raises issues that have been discussed for years stating “lack of comfort” with the proposal, suggesting alternative approaches, and demanding more time to consider or reject. At this point, everybody unhappy with compromises made along the way chirps in with counter-points made over the years and the proposal is either withdrawn or defeated by a 20% minority, many of whom did not take part in previous discussions. We think that

- “lack of comfort” is not sufficient to block a proposal

Even when (as is common) there are minor remaining issues, typically, there are many months between a proposal being approved (by any part of the committee) and the final standard during which non-fatal flaws can be fixed. We think that a proposal that has passed a committee (say EWG or LEWG) should be accepted without undue delays by further groups. Obviously, new inputs should be considered, but in the absence of new information, “we considered and discussed that” by a WG chair should be conclusive:

- People who were not sufficiently motivated to take part in the discussion should feel obliged to at least stay neutral. Similarly, if your position has been discussed and voted down in WG, then you should at least vote neutral in plenary, to at least abide by the group’s position.

Unfortunately, this principle is easily gamed by people who break it to block proposals they don’t like. This has been observed in the committee and has led to resentment and erosion of trust.

At a first approximation,

- No proposal is ever perfect.
- Different people have very different ideas of what degree of perfection is needed for a proposal to be part of the standard (or part of a TS).

“The last bug” is a common programmer’s joke and our language specification reads like a program in a poorly defined language (English) without advanced control structures, without abstraction mechanisms, and without a compiler. Consequently, perfection will always elude us and we should take that into account when we decide what is “good enough” for a standard (or a TS).

We have seen proposals move forward with issues lists to be considered before a final vote or even after. We strongly encourage that approach to make progress

- If a proposal is fundamentally sound, it should be moved forward, even into the WP text
- Refinement of text can (and should) happen later
- There are problems that are unlikely to be found until after the wording for a proposal is integrated into the WP text (as in software development: “integrate early to allow testing”)
- Addition of desired but incompletely-developed features can wait until later, even to a later standard
- As the shipping date for a standard approaches, the efforts to “polish” the text and drain the issue lists should be redoubled at the expense of effort on new proposals (as is currently done)

Through delays, a good proposal can get dated, failing to benefit from years of improvements and progress in the community. Note the file system TS and networking TS each has a decade of use behind them. Similarly, **std::variant**, **std::optional**, and **std::any** have a long history as independent proposals. That wouldn’t be too bad if the reason was that significant improvements were added during the process. However, the current process tends to reduce proposals to their most conservative cores. Thus, some “novel” C++ features feel dated by the time they are accepted.

- Aim for prompt delivery followed by incremental improvements

Note that this approach can only succeed if the end-goals are reasonably clear.

- Articulate the end goals for a proposal

When triaging features for consideration, we propose that the WG chairs to put a higher priority on features that conform to these goals, but also keep a watch against features that

- Turn C++ into a radically different language
- Turn parts of C++ into a much significantly different language by providing a segregated sub-language
- Have C++ compete with every other language by adding as many as possible of their features
- Incrementally modify C++ to support a whole new “paradigm” without articulating the end goal
- Hamper C++'s use for the most demanding systems programming tasks

- Increase the complexity of C++ use for the 99% for the benefit of the 1% (us and our best friends.)

The committee members and proposal authors are a group of volunteers who may have different aims, different views on industrial practice, different views of necessary skills, or different directives from the organization they represent. As we propose features, we urge the group to maintain cordial discourse, and aim for what is best for C++ as a language and not merely what is best for your company's current direction. This sometimes means willing collaboration with people with different aims and priorities. For the most part, we have been successful. More specifically, don't oppose a proposal just because:

- it is seen as competing with your favorite proposal for time/resources
- it is not relevant to your current job
- you have not personally reviewed it
- it is not perfect (according to your principles)
- it is not coming from your friends
- it is coming from someone you have been at odds with on different subject

Remember, we are writing a standard for millions of programmers to rely on for decades, a bit of humility is in order [Stroustrup,2019b].

## 7.2 Proposal processing

WG21 does not lack for proposals. The interest is high and the number of proposals of each mailing has grown to more than 100. That's unmanageable for an individual with a day job. To stay focused on the priorities discussed in this document, we recommend that chairs focus at least 60% of their time on advancing these priorities. How that is done is entirely the chairs prerogative. For example:

- Raise the barrier for repeated presentation of and voting on failed proposals.
- Require a clearly specified written rationale for each proposal.
- Require a short written tutorial for each proposal.

For any non-trivial proposal, it is not obvious how to use the new facility well in combination with other features. It is easy to say "teaching this is easy" but such statements should be backed-up with examples and experience report (or be clearly marked as conjecture). In particular, for whom is learning and using a new feature easy?

- Never assume that the use of a proposed feature is easy and obvious to everybody.
- Never assume that the need for a proposed feature is obvious to everybody.

The volume of proposals is such that people are not able to track all upcoming votes. This has led to "no" votes in WGs and plenary. We ask the committee to consider process improvements that improve this communication and notification, but also ask that members actively review and discuss with others their proposal progress, so that the burden is fairly spread. At one time, Alistair Meredith did a great job of summarizing the flow and progress of each paper prior to C++11. Today this is unmanageable for one

person. We have noted other forms of process improvement that has helped improved communication, and understanding the issues as we review papers. These include use of githubs, and summary notes recorded for each proposal, to save readers the bother of going through each meeting minutes to trace the history of a proposal and find the related papers. We do not necessarily enforce any particular methods, but feel some method should be considered to allow other groups to track progress:

- WGs and SGs should maintain an up-to-date brief list of the status of proposals being processed.
- WGs and SGs should - as far as possibly - announce what proposals will be up for discussion

Such lists should be easily accessible. It is not good enough to have information aimed at keeping people who are not regular members of a WG or SG posted on a hard-to-find mailing list requiring separate signup.

Remember the need for trust (§6.1):

- “I didn’t have time to read the document” is not sufficient reason to oppose (provided the paper was submitted on time)
- “I don’t understand” is not by itself sufficient reason to oppose (give reasons and examples why your lack of understanding is not just your own problem)

### 7.2.1 Guidance on Converging on unified proposals

In our experience, nearly all major proposals and a few medium proposals since C++11 have triggered alternative proposals. These lead to divergent consensus. In some cases, an alternative proposal has been brought forward at the last moment, only because it sometimes take that long for people to understand the original proposal, to have a partial implementation of the original, see the other possibilities, or, very occasionally, it just takes that long to take notice. Sometimes, these alternative proposals have come with riders for scope creep.

We have seen this time and time again, e.g., with Concepts, Executors, and Contracts. Undoubtedly, we will see it again with future proposals. This is a good sign because it is an indication that the community cares deeply enough to want the best, and to have it represent the broadest community possible. In the following guidance, we are specifically referring to design differences, rather than implementation differences, though we accept some amount of implementation consideration needs to be factored in.

The problem of ratification of the first iteration usually comes as to when is the scope sufficient for a first design to be added to the Standard when there are divergent design directions:

- Which of several relatively attractive design proposals to choose and why?
- Which extra feature to add and why?
- What features of the status quo should be removed and why?
- Which feature should be modified and why?

Admittedly convergence is hard, especially in an emotionally charged environment, with ratification deadline bearing down, where everyone now has a huge stake. We can only ask that stakeholders try hard to adhere to the following as we have observed it to work in the past because stakeholders have

come with a willingness for honesty, compromise, wish for progress, a unified goal and a long-term realistic future perception, lacking ultimatums. Though it may be obvious, it is still worth stating:

- Aim for minimal intersection or union set that at least satisfy as many major stakeholders as possible. But do not aim for all possible stakeholders as that is untenable.
- Once you have competing proposals, work hard to get a minimal starting set, instead of building everything into the first iteration. This is surprisingly hard and should not be underestimated.
- Resist the urge to add a favourite rider feature in the initial iteration, especially when the door seems partly open to inject a significant change. This demands a great deal of discipline.
- Authors of competing proposals should get together to write a joint paper (or papers) comparing the proposals with key usage examples and key implementation issues.
- Always think in the long term, that this initial entry into the Standard is just there to gain initial experience with a minimal set, but know that if successful there is room for addition. Probably the best example of that is `constexpr`.

Examples of successful iteration improvements that we urge you to use as template in seeing a long-term view of major proposals and how we build on previous work follows. In all cases, we have found a collaborative and common vision helps to soothe individual egos.

- Templates (simple non-member templates -> member templates -> additional template kinds -> concepts)
- Lambdas (non-polymorphic -> polymorphic -> constant-evaluated -> ? recursive),
- `constexpr` (single return statement -> general statements -> dynamic allocation & more -> ? reflection).
- In many ways this also applies to library components (e.g., concurrency facilities).

Note that some differences are irreconcilable. In such cases, the committee has to make choices. We are not asking people to create bloated messes just to plaster over fundamental differences. C++ cannot be all things to all people.

### 7.2.2 ABI Stability

We believe that ABI breakage, both in the language and the library, should be considered on the merits of each individual proposal which necessitates such a break. We do not believe the labeling of any specific C++ standard as “your chance to break ABI” to be healthy for the future of C++. Such a label would encourage ABI breaks which may not bring a correspondingly high benefits to offset the cost of said ABI break. [P1654](#) contains some additional questions, observations and summaries of past ABI breaks (see, in particular, “Section 3: Past ABI breaks”).

The ABI Review Group (ARG) has been set up to help analyse the potential impact of ABI breaking proposals. Given the impact ABI changes can have on various implementations, such a board is in the best position to evaluate the potential consequences of such changes and the effort required to mitigate them or transition across those changes. The board makes recommendations, and as a group it also prevents any one single implementation from over-weighing their influence. As earlier versions of this paper suggested the board therefore includes at most two representatives for any product/company



(one for the core language and/or one for the standard library); that does not exclude input from other people, of course: representatives are encouraged to consult with the experts they know.

### 7.2.3 Safety and Security

We supported the creation of a Safety and Security Review Group (SSRG) to serve our directions in Section 4 Long Term Aims (Decades) but it can only review proposals with safety and security in mind. I.e., like the ABI Review Group (ARG) this SSRG group is only advisory.

### 7.2.4 C/C++ Liaison

Similarly, we were instrumental in motivating a C/C++ Liaison SG (SG22 on the WG21 side) to improve on C and C++ cross compatibility and actively review existing and future proposals with such aim. This group has a presence in both WG21 and WG14 to avoid repeating some infelicitous divergences of the past (e.g., the incompatibility between C and C++ “inline function” mechanisms).

## 7.3 The role of TSs

Technical Specifications (TSs) have become popular. In theory, they provide an intermediate stage where major new features can be discussed, specified, tested under fewer constraints than for the standard itself, and later be moved into the standard with changes based on the experiences gained. The move from a TS to an IS is supposedly simplified by the significant work to complete the TS and the experience gained.

In practice, this seems to work reasonably well for libraries, though we see examples where facilities are stripped from a proposal as being too advanced, rather than pushed forward for experimentation. For language features, the experience is less positive. It seems that the barrier to entry into a TS is not significantly lower than for an IS, and the effort devoted to complete a TS detracts from experimentation, freezing the language feature in time, years before acceptance. After the TS, most of the design and specification issues are then revisited a second time. Thus, a TS becomes a method for delaying a proposal. Also, a TS doesn't seem to be sufficient to encourage multiple implementations. This implies that large parts of the C++ community doesn't get to use the facility, large organizations using multiple compilers can't experiment with the TS at scale, and tool builders hold back waiting for a “proper standard” supported by all major implementers. Consequently, detractors can dismiss the experience reports and clamor for novel alternatives.

We recommend

- Use TSs for library components.
- Don't use TSs for a language feature unless the feature is a mostly self-contained unit.
- Never use a TS simply to delay; it doesn't simplify later decision making.
- When proposing a TS, specify the “aim”: what the TS is supposed to learn or achieve.
- List “exit criteria” (TS to IS or whatever target) to allow people to determine whether the work is complete and whether it succeeded.

- Consider other vehicles such as SG (Study Groups), IS, and not just TS
- Consider some or all the following incomplete list of frequently asked questions in your deliberations and TS proposal and record their answers along with the aim and exit criteria:
  - Is there an implementation?
  - Is it a Library or Language proposal, or does it involve both aspects?
  - Is the proposal a foundational proposal, meaning many other C++ aspects/proposal depend on it, and/or it depends on many other C++ aspects/proposals?
  - Is it independent of aspects of the language?
  - Are there competing design proposals?
  - Is the proposal complicated or large that you fear there will be error in design decision?
  - Is it a research idea?
  - Is there substantial invention?
  - Can it be staged?
  - Is there a subpart that deserves to be in IS?
  - Is the wording complicated or unconventional?
  - Will the proposal benefit from early integration (can be applied to a WP)?
  - Will you get feedback/testing only after TS publication or IS publication?
  - Is there a motivation to slow down a proposal?
  - What would it take to turn the TS into an IS?
  - Are you juggling a large number of related or dependent proposals (other proposals that depend on this proposal)?
  - Are you aiming for user feedback?
  - Are you aiming for implementation feedback?
  - Is there a scheduling concern to make C++xx for it or its dependents?

We have found some or all these questions are always asked anyway at each SG deliberation, so proposers should consider them early on. Expect the DG to offer non-binding advisory (TS or IS or other target) in some cases and we hope you weigh our opinion as part of your decision process.

If a proposal isn't ready for the standard let it be improved or rejected in the appropriate WG or SG (Study Group). Don't add a formal TS process.

## 7.4 “Details” vs. Design

The committee spends most of its time on details that the average programmer will never notice. A lot of that is necessary. However, it seems that most members spend most of their time on such details during the design phase; that’s wasteful and weakens the language as experienced by users. Instead, early on in the process, try to spend the majority of time on design issues, such as:

- What problems are this feature meant to solve? (D&E recommends not to accept a feature that doesn’t solve two apparently unrelated problems)
- What alternative solutions are possible?
- What related features should be considered simultaneously with this one?
- How well does this feature fit with the general style of the language and standard library?
- Is this feature aimed only for a few language or library experts or will it be used in application code?

An individual proposal is essentially never useful in isolation; it will work in combination with other features:

- Consider every feature in the context of features with which it will be used
- Don’t approve a set of related features one by one, evaluate them together

In particular, cluster related proposals (language and library) together.

- Be fair when comparing alternative proposals

In particular, try hard to give discussions of advantages and disadvantages equal weight for each alternative. Presenting only advantages for a “favored proposal” and only “disadvantages” for an unfavored alternative is not acceptable.

During the design phase, write and maintain a tutorial for the feature. It will help others understand what the feature is supposed to achieve and help keep features aimed at “ordinary programmers” from drifting into expert-only territory. A good tutorial usually goes from the simple towards the more advanced and from the concrete to the more abstract; the tutorial in K&R is an excellent example.

The text of the standard should be precise and comprehensive. It is not supposed to be a tutorial, but the ideal is that after some “acclimatization” an experienced programmer should be able to interpret the text with some confidence. This is not currently the case. When crafting WP text

- Consider readability by people not members of the WG crafting the text
- Non-normative notes can be useful
- Short examples can be very useful
- Try not to change the meaning of terms common in the C++ community
- Where possible, use terms common in the C++ community

Remember that compiler writers and standard-library implementers are not the only target audience for the standard (or a TS).

## 7.5 Rationale, Design, and Wording

We have a standing document on how to submit a proposal:

<https://isocpp.org/std/submit-a-proposal>

which in turn links to a paper from 2012:

<http://open-std.org/jtc1/sc22/wg21/docs/papers/2012/n3370.html>

These two papers form great starting point on what a good initial proposal should look like. However, we have observed that the rationale and design parts are often left out of later documents containing wording and often not maintained to reflect decisions during the time taken to process the proposal. This time is typically multiple years and covers multiple meetings in multiple WGs. This can be a significant barrier to understanding and a source of confusion. For example “Is the wording in Pxxxx supposed to reflect the design in Pyyyy?” How would I know? It can be significant work finding out, often involving searching through the wikis for multiple meetings. Non-members of the committee cannot do that, so they really can’t understand what is being proposed.

It has become popular to start papers with a change record. This is welcome and useful, but not sufficient because applying multiple deltas - and often deltas to deltas - to a document requires significant effort and can be a source of misunderstanding.

We therefore suggest that the rationale and design discussions from early papers are carried through and maintained in later wording documents. This makes documents larger, but people can skip what they don’t feel the need to read. Having all in a single document also provides redundancy that can help spot mistakes. It should also help decrease the confusion that often happens when a proposal comes up for final vote so that people who have not been following the proposal closely have to catch up.

## 7.6 Balance of concerns and engineering approach tradeoffs

Successful language design - like all successful engineering - requires fundamental good ideas and balancing constraints. Optimizing for one thing - in one direction - may succeed for one application area for one moment of time, but eventually the result dies for lack of adaptability. In our case, focusing exclusively on performance deprives us of new entrants into the community and the new ideas they bring. It also leaves us wide open to becoming blindsided by changes in hardware or optimization techniques. Super-optimized code tends to run poorly on next year’s architecture and only run well on the kind of tasks for which it is optimized. This is particularly true if language facilities are modified to support specific optimizations. Furthermore, optimizations are often devoid of fundamental good ideas; they focus on tuning some status quo.

On the other hand, optimizing exclusively for ease of use can lead to very slow code. That must be avoided. There are a variety of techniques for dealing with the elegance vs. performance dilemma. One is the onion principle: there is an elegant, reasonably performant, and usually type safe “layer” that most people can use most of the time. When more performance or more special cases need to be

handled, we peel off a layer and gain more control, more performance for specific cases, and more opportunities for errors. If that's not enough, we peel off yet another layer. This can go on until you directly manipulate specific hardware features. The reason for referring to that as the onion principle is that "each time you peel off a layer you cry more." Another technique for gaining both performance and simplicity is to replace standard-library components with compatible or close-to-compatible implementations tuned for a specific set of uses.

To serve its community well and thrive, C++ must address the real problems of a huge community, rather than obsess over extreme demands of a relatively small group of experts. Performance is important, but it cannot be the only concern of the community: simplicity, safety, toolability, ease of teaching, maintainability, composability of software from different sources, compilation speed, and stability are other essential concerns. There are many ways of improving performance, including cleaner code, coding guidelines, optimization hints, specialized implementations of standard components, specialized libraries, optimizers tuned for specific hardware, and specialized ABI for specific application areas.

C++ has survived for 40 years by carefully balancing concerns, learning from experience, and avoiding chasing fashions. It started up with two aims:

- utilize hardware well
- offer effective abstraction mechanisms

When given a choice between adding language features to address a problem directly and improving the abstraction mechanism, the bias was in favor of the latter. This served us well as hardware and programming techniques evolved. Furthermore, there is a constant tension between

- evolve to better address novel challenges
- stability/compatibility

When given that choice, the choice almost always was to do both. That leads to a large, more unwieldy language, but it doesn't cut users off by the knees, forcing them to choose between adopting new languages (always tempting) and avoiding costly reworking of their code.

Every language has made different engineering tradeoffs, but few have succeeded in C++'s core domains. Those language design trade-offs are engineering choices in portability vs performance vs productivity. We do not focus on any one of these to the exclusion of all others.

The nirvana of programming, one that offers world-class performance, unmatched productivity, full generality, **complete safety, and ease of writing**, simply does not exist. Until such a nirvana appears, it will be necessary to make engineering tradeoffs among performance, productivity, and portability.

There are a number of items that may be involved with any proposal that interacts with existing practice, such as an extension provided by one or more vendors. Firstly, consider how well the item fits into the C++ programming model. Adding inconsistent behavior would make it hard to use such a feature easily and safely. Secondly, does the proposal completely match the existing practice or are there differences? Where a similar feature is available from multiple vendors this can be particularly troubling. If there are differences, it may be better to deliberately choose a different name or syntax

to avoid the risk of causing confusion (this was done, for example, with "unordered\_set" because of the potential confusion with pre-existing "hash\_set" extensions.)

## 7.7 C++ future online and F2F

The current pandemic may last for many more months and possibly pandemics will become more frequent, so we should adapt to a possibly hybrid development model of being online and/or F2F. The committee has pivoted extremely quickly with multiple SGs coming online as we face a rolling ban to F2F meetings which seems to expand continuously.

There are some cautionary observations:

- Many members are overwhelmed as their work (and life in general) is complicated by lock-downs and the variety of organizations they are part of tries to maintain progress by adding more meetings.
- Each week there are many WG21-created (WG and SG) meetings. Few if any can attend them all and many can't consistently attend all meetings of groups they traditionally attend in person.
- The on-line meetings have a changing sets of attendees, often with proponents of new features turning up in large numbers. Often traditional "stalwarts" of a group are missing.
- For some, the steady stream of "mailings" and meetings are hard to keep up with as they disperse effort among other commitments. It can be very hard to disengage from "day job" tasks.
- The WG21 mailing lists alone produce in total between 1,000 and 3,000 messages a month, which is a large volume of traffic to keep on top of.
- The week-long face-to-face meetings allowed people to concentrate on WG21 progress — sometimes across WGs and SGs. Such concentration can be hard to achieve for many short specialized meetings. There is no time for "the cache to be loaded."

We are concerned that this train has cars going at different speeds in many directions: it is hard to maintain a direction when we cannot meet F2F. Many short meetings favor progress on smaller, easier to isolate, features over major features with broad impact on the language and/or standard library.

We should be careful when we change our processes. People can't keep up when the content, logistics and process are changing very fast. We should also be very careful not to critically rely on processes and tools that some people are not familiar with and possibly cannot utilize from their work (or home) locations. There is a bewildering variety of on-line collaboration tools, video-conferencing systems, IMs, calendar systems, etc. Many would prefer to work on C++ than learning and keeping up-to-date with a shifting set of collaboration facilities and conventions.

## 8 The C++ Programmers' Bill of Rights

This note was posted to reflectors, presented at the June 2017 Toronto meeting and discussed. We propose that it be formally adopted by WG21:

## 8.1 “The C++ Programmers’ Bill of Rights.”

We, the ISO C++ Standards Committee, promise to deliver the following to the best of our ability, assuming user code adheres to the current standard:

1. **Compile-time stability:** Every significant change in behavior in a new version of the standard is detectable by a compiler for the previous version.
2. **Link-time stability:** ABI breakage is avoided except in rare cases, which will be well documented and supported by a written rationale.
3. **Compiler performance stability:** Changes will not imply significant added compile-time costs for existing code.
4. **Run-time Performance stability:** Changes will not imply significant added run-time costs to existing code.
5. **Progress:** Every revision of the standard will offer improved support for some significant programming activity or community.
6. **Simplicity:** Every revision of the standard will offer simplification for some significant programming activity.
7. **Timeliness:** Every revision of the standard will be shipped on time according to a published schedule.

Note “to the best of our abilities”. These are ideals or guiding principles, rather than executable statements. For example, if a function is added to a header file, the compilation of code that includes that header will slow down imperceptibly. That’s acceptable. Adding enormous amounts of code to a header so that compilation slows noticeably would be another matter.

Note that the “improved support” in “improved support for some significant programming activity or community” can be just about anything, such as a language feature, a standard-library, a hook for tool support, a generalization of an existing set of features. It is not our aim to destabilize the language with a demand of constant change; rather to help the committee focus on what is significant to the community as opposed to insignificant changes and churn.

These are ideals. They are what we would like to see done. If we succeed, most users will be very happy. However, they are not a recipe we could blindly follow to deliver a new standard. As is typical for ideals, they can conflict: there are tensions among the desired goals. This is common: Ideally, we want quality, on-time delivery, and low cost of products, but we know from experience that it is very hard to get all three. We want freedom of speech and absence of verbal intimidation, but balancing those two can be very hard. It is the same for the ideals of the “The C++ Bill of Rights”; we want all, but the committee will have to make hard choices.

These are ideals. They are meant to be rather grand statements, rather than nit-picked long sentences watered down by adjectives and caveats. It will be up to the committee members to interpret the ideals in the light of real design and scheduling problems.

There are just seven ideals listed and they are quite general. We could add many more, but a laundry list of specifics would dull the appeal and be impossible to remember in the heat of discussions about direction, what can be delivered when, and specific technical concerns.

## 9 Caveat

The members of the direction group have been members of the standards committee for decades. We were (we think) chosen because we have deep and broad experience with the C++ technology, use, and community. We all have well-documented track records. In the role of members of the direction group, we try to serve the good of the C++ community as a whole. It is not possible to completely separate what we consider best for the C++ community from the specific proposals we work on, but we try not to unfairly favor our own proposals. Similarly, we try not to discourage proposals or actions just based on personal biases. Some of our recommendations may not be universally appreciated, but nothing we say is meant to insult anyone.

Some of the recommendations here could be seen as contradictory. We see such as fundamental tensions in a design, requiring tradeoffs, rather than contradictions. Design is hard, design by committees is even harder.

## 10 Acknowledgements

Many of the arguments and points of view have a long history in the committee. Thanks to all who contributed. In particular, thanks to the authors of the documents we reference here.

## 11 References

- [Bastien,2018] J.F. Bastien: Feedback on 2D Graphic. P1225R0. 2018-10-02.
- [Berne,2019] Joshua Berne, Timur Doumler, Andrzej Krzemiński, Ryan McDougall, Herb Sutter: [Contracts — Use Cases](#). P1995R0. 2019-11-22.
- [CG,2015] [C++ Core Guidelines](#).
- [Clow,2018] Marshall Clow, Beman Dawes, Gabriel Dos Reis, Stephan T. Lavavej, Billy O’Neal, Bjarne Stroustrup, and Jonathan Wakely: Standard Library Modules. P0581R1. 2018-2-6.
- [GDR,2016] G. Dos Reis, J. D. Garcia, J. Lakos, A. Meredith, N. Myers, B. Stroustrup: [A Contract Design](#). P0380R1. 2016-07-11.
- [GDR,2017] G. Dos Reis, J. D. Garcia, J. Lakos, A. Meredith, N. Myers, B. Stroustrup: [Support for contract based programming in C++](#). P0542R2. 2017-11-26.
- [Nishanov,2017] Gor Nishanov: [C++ Extensions for Coroutines TS Document](#). N4680. 2017-07-30.
- [Hinnant,2017] Howard Hinnant: [Extending <chrono> to Calendars and Time Zones](#). P0355R4. 2017-10-1.
- [McLaughlin,2017] Michael McLaughlin, Herb Sutter, Jason Zink, Guy Davidson: [A Proposal to Add 2D Graphics Rendering and Display to C++](#). P0267r6. 2017-07-30.



- [P1654] ABI WIP document. 2019-06-17.
- [Quora1] [Can I use C++ for hard real-time embedded systems and kernel development?](#)
- [Quora2] [Does C-11 and C-14 make it easy to use -C++ in kernel development?](#)
- [Saks,2016] Dan Saks “extern c: Talking to C Programmers about C++”. CppCon 2016 keynote.
- [SD-8] [SD-8: Standard Library Compatibility](#)
- [StdLibGuidelines] [Standard Library Guidelines](#)
- [Stroustrup,1994] B. Stroustrup: The Design and Evolution of C++. Addison-Wesley. 1994.
- [Stroustrup,1993] B. Stroustrup: [A History of C++: 1979-1991](#). Proc. ACM HOPL-2. 1993.
- [Stroustrup,2007] B. Stroustrup: [Evolving a language in and for the real world: C++ 1991-2006](#). Proc. ACM [HOPL-III](#). 2007.
- [Stroustrup,2014] B. Stroustrup: [Programming -- Principles and Practice Using C++ \(Second Edition\)](#). May 2014. Addison-Wesley. ISBN 978-0321992789. 1<sup>st</sup> edition 2008.
- [Stroustrup,2015] B. Stroustrup: [Thoughts about C++17](#). N4492. 2015-05-15.
- [Stroustrup,2015b] B. Stroustrup, H. Sutter, and G. Dos Reis: [A brief introduction to C++'s model for type- and resource-safety](#). Isocpp.org. October 2015. Revised December 2015.
- [Stroustrup,2018] B. Stroustrup: [NDC Keynote: What can C++ do for embedded systems developers?](#)
- [Stroustrup,2018b] B. Stroustrup: [Remember the Vasa!](#) P0997R0.
- [Stroustrup,2018c] B. Stroustrup: [The Evils of Paradigms](#). P0176R0.
- [Stroustrup,2019a] B. Stroustrup: [C++ exceptions and alternatives](#). P1947. 2019-11-18.
- [Stroustrup,2019b] B. Stroustrup: [How can you be so certain?](#) P1962R0. 2019-11-18.
- [Stroustrup,2020] B. Stroustrup: [Thriving in a crowded and changing world: C++ 2006-2020](#). P2184R0. 2020-06-10.
- [Stroustrup,2022] B. Stroustrup and G. Dos Reis: Design Alternatives for Type-and-Resource Safe C++. P????R0. 2022-10-15.
- [Sutter,2018] [WG21 Practices and Procedures](#). ISO/IEC JTC1/SC22/WG21/SD-4. 2018-01-17.
- [Winkel,2017] JC van Winkel, Jose Daniel Garcia, Ville Voutilainen, Roger Orr, Michael Wong, and Sylvain Bonnal: [Operating principles for C++](#). P0559R0. 2017-01-31.
- [Voutilainen,2017] Ville Voutilainen: [To boldly suggest an overall plan for C++20](#). P0592R0. 2017-02-05.
- [Wakely,2017] Jonathan Wakely: [Extensions for Networking](#). N471. 2017-11-27.

- [Winters,2017] Titus Winters, Bjarne Stroustrup, Daveed Vandevoorde, Beman Dawes, Michael Wong, and Howard Hinnant: [C++ Stability, Velocity, and Deployment Plans](#). P0684R0 2017-06-19.